

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 23 ✓



By- Vishal Sir



Recap of Previous Lecture



Topic

Finite group

Topic

Addition modulo 'm' $\oplus_m$

Topic

Multiplication modulo 'm' $\otimes_m$

Topic

Order of an element in a group $(G, *)$

Topics to be Covered



Topic

Subgroup



Topic

Cyclic group





Topic : Finite Group



- A group $(G, *)$ with finite number of elements in set G is called a finite group.
- If $(G, *)$ is a finite group, then $|G|$ is defined as order of group $(G, *)$.
denoted by $O(G)$ or simply $|G|$



Topic : Finite Group



eg. The set $\{0\}$ is a finite group of order = 1
with respect to binary opⁿ addition

eg. The set $\{1\}$ is a finite group of order = 1
with respect to binary opⁿ multiplication.

eg. The set $\{1, -1\}$ is a finite group of order = 2
with respect to binary opⁿ multiplication

✓ Above three groups are the only finite groups of
real numbers w.r.t. binary opⁿ addition and/or Multiplication

Note:-

- ✓ ① If "e" is the identity element w.r.t. binary opⁿ "*", then the set containing element "e" only $\{ \underline{e}, \{e\} \}$ will always form a finite group of order = 1, w.r.t. binary opⁿ "*".
- ② In a finite group of order = 2, every element is inverse of itself.



Topic : NOTE

Let "S" is any non-empty set,

"S" with respect to binary opⁿ "*" is a group

if and only if, $\textcircled{*} \textcircled{1} \quad \underline{a * b^{-1} \in S, \forall a, b \in S}$

and $\textcircled{2} \quad \text{"*"} \text{ is associative}$

it should be
enough to
Prove that

- $\textcircled{1}$ S is closed w.r.t. "*"
- $\textcircled{2}$ Identity w.r.t. * belongs to S
- $\textcircled{3}$ Inverse for every element of set S is present in S.

Let

$$a * b^{-1} \in S, \forall a, b \in S$$



(Identity)

① Prove Identity $e \in S$

Let $a \in S$

i.e. $a, a \in S$

We know

if $a, a \in S$, then $a * a^{-1} \in S$

(identity) $e \in S$

if there exists any element
in set S ,
then identity $e \in S$

(Inverse)

② Prove if
 $a \in S$, then $a^{-1} \in S$

(i) we just proved $e \in S$
if e is the only element
in the set, then we know
 $\text{inv}(e) = e \therefore$ inverse

exist
→ if there exists any other
element $a \in S$,
then we know $e, a \in S$

if $e, a \in S$, then $e * a^{-1} \in S$
 $\therefore a^{-1} \in S$

i.e. inverse exist for
every element $a \in S$

(Closed)

③ Prove
if $a, b \in S$, then $a * b \in S$

Let $a, b \in S$

We just proved

if $a, b \in S$, then $a^{-1}, b^{-1} \in S$

* i.e. if $a, b \in S$
then $a, b^{-1} \in S$

→ and hence

$a * (b^{-1})^{-1} \in S$

$a * b \in S$

i.e. Closure property
is also satisfied

#Q. Which of the following statement is/are not true.

- A** $\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer (True) ✓
- B** $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication ✓
 it is a monoid but not a group $\text{inv}(0.5) = 2 \notin \text{Set} \therefore \text{Not a group}$
- C** $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication
 group ✓ (True) $2^{n_1} * 2^{n_2} = (2)^{n_1+n_2} \therefore \text{Closed}$
 $2^0 \in \text{Set} \therefore \text{identity exists}$
- D** None of these



Topic : Finite Group

Cube roots of unity are $1, \omega, \omega^2$

$$\omega^3 = 1$$

The set $\{1, \omega, \omega^2\}$ form a finite group of order = 3
w.r.t. multiplication.

- ① Closed ✓
- ② Associative ✓
- ③ Identity is present ✓
- ④ Inverse ✓

To check for inverse
We can use Composition
table

Multiplication

1	ω	ω^2
1	ω	ω^2
ω	ω^2	$\omega^3 = 1 = e$
ω^2	$\omega^3 = 1 = e$	$\omega^4 = \omega$



Topic : Finite Group

Four roots of unity are $1, -1, i, -i$
 $i^2 = -1$ & $i^4 = 1$



The set $\{1, -1, i, -i\}$ form a finite group of order = 4
w.r.t. multiplication.

- ① Closed ✓
- ② Associative ✓
- ③ Identity is present ✓
- ④ Inverse ✓

$\text{inv}(1) = 1$
 $\text{inv}(-1) = -1$
 $\text{inv}(i) = -i$
 $\text{inv}(-i) = i$

	1	-1	i	-i
1	$1=e$	-1	i	-i
-1	-1	$1=e$	-i	i
i	i	-i	$i^2 = -1$	$-i^2 = -(-1) = 1=e$
-i	-i	i	$-i^2 = 1=e$	$i^2 = -1$

A set of 'n' roots of unity will form a finite group
of order = n w.r.t. multiplication



Topic : Addition modulo ' m ' $\oplus_m$

it gives remainder

Let m is any fixed positive integer
for any two non-negative integers a & b ,

$$a \oplus_m b = \begin{cases} (a+b) & , \text{ if } (a+b) < m \end{cases}$$

|||
 $(a+b) \bmod m$

r

otherwise

Where ' r ' is the remainder
obtained on dividing $(a+b)$ by m

$$2 \oplus_6 3 = 5$$

$$4 \oplus_6 5 = 3$$



Topic : Multiplication modulo 'm' $\otimes_m$

Let m be any fixed positive integer,
for any two non negative integers a & b ,

$$a \otimes_m b = \begin{cases} (a \cdot b) & \text{if } a \cdot b < m \\ \pi & \text{otherwise} \end{cases}$$

|||
($a \cdot b$) Mod m

otherwise

Where π is the remainder
Obtained on dividing ($a \cdot b$) by m .

$$\begin{aligned} 2 \otimes_7 3 &= 6 \\ 4 \otimes_7 5 &= 6 \end{aligned}$$



Topic : NOTE

★

① The set $\{0, 1, 2, \dots, (n-1)\}$ form a finite group of "order = n " w.r.t. binary opⁿ $\oplus_n$.

eg. The set $\{0, 1, 2, 3, 4\}$ form a finite group of order = 5, w.r.t. $\oplus_5$.

$$\begin{aligned} \text{inv}(0) &= 0 \\ \text{inv}(1) &= 5-1=4 \\ &\Rightarrow 1 \oplus_5 4 = 0 \end{aligned}$$

$$\begin{aligned} \text{inv}(2) &= 5-2=3 \\ &\Rightarrow 2 \oplus_5 3 = 0 \end{aligned}$$

$$\begin{aligned} \text{inv}(3) &= 2 \\ \text{inv}(4) &= 1 \end{aligned}$$

① Set is closed w.r.t. $\oplus_5$

② $\oplus_n$ is associative for any +ve integer 'n'

③ Identity element w.r.t. $\oplus_n$ is always '0'.

④ Inverse: $\text{inv}(0) = 0$ {because '0' is identity}

for any other element $x \in \text{Set}$
 $\text{inv}(x) = (n-x)$ w.r.t. $\oplus_n$

Note:

Let

$$A = \{x \mid x \in \overset{\text{Natural numbers}}{N} \text{ and } x < \overset{\text{given fixed positive integer}}{n} \text{ and } \text{GCD}(x, n) = 1\}$$

$\rightarrow$ A is a set of all natural numbers that are less than ' n ' and Coprime to ' n '

let $n = 15$,

then

$$A = \{1, 2, 4, 7, 8, 11, 13, 14\}$$

Set of all natural numbers which are less than 15 & Coprime to 15.

If GCD of x & n is '1' then x & n are called Co-prime numbers (or) relatively prime numbers



Topic : NOTE

② The set of all natural numbers that are less than n and Co-prime to n , form a group w.r.t. $\otimes_n$

eg: let $n=8$, then

$\{1, 3, 5, 7\}$ will form a finite group of order = 4 w.r.t. binary opn $\otimes_8$.

Set of all natural nos that are less than 8, & Co-prime to 8.

→ $\{1, 3, 5, 7\}$ is a finite group of order = 4,
w.r.t. $\otimes_8$.

① Closed. ✓

② $\otimes_n$ is associative
for any +ve integer 'n'. ✓

③ '1' is the identity element
w.r.t. $\otimes_n$

④ Inverse:
 $\text{inv}(1) = 1$
 $\text{inv}(3) = 3$
 $\text{inv}(5) = 5$
 $\text{inv}(7) = 7$

$$1 \otimes_8 1 = 1 \text{ (identity)}$$

$$3 \otimes_8 3 = 1 \text{ (identity)}$$

$$5 \otimes_8 5 = 1 \text{ (identity)}$$

$$7 \otimes_8 7 = 1 \text{ (identity)}$$

it just happened that each element
is inverse of itself, it is not
necessary for all such groups.

eg. Let $n=15$, then
 $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a finite group of order = 8 w.r.t. $\otimes_{15}$

Find inverse of every element of the group.

identity element = 1

$$\text{inv}(1) = 1$$

$$\text{inv}(2) = 8, \Rightarrow \text{inv}(8) = 2$$

$$\text{inv}(4) = 4$$

$$\text{inv}(7) = 13 \Rightarrow \text{inv}(13) = 7$$

$$\text{inv}(11) = 11$$

$$\text{inv}(14) = 14$$

Note: ① $\{1, 3, 5, 7\}$ is a group w.r.t. $\otimes_8$

② $\{1, 3\}$ is also a group w.r.t. $\otimes_8$

③ $\{1, 5\}$ is also a group w.r.t. $\otimes_8$

④ $\{1, 7\}$ is also a group w.r.t. $\otimes_8$

→ Some subset of original group may also be group

⑤ $\{1, 3, 5\}$ is not a group w.r.t. $\otimes_8$

→ i.e. Every subset need not be a group



Topic : NOTE

③ If p is a prime number, then
 $\{1, 2, 3, \dots, p-1\}$ will form a group of
order $= (p-1)$ w.r.t. binary opⁿ $\otimes_p$

eg. $\{1, 2, 3, 4, 5, 6\}$ is a group of
order $= 6$, w.r.t. opⁿ $\otimes_7$

Any no. less than p
will be Co-prime to p ,
 $\therefore \{1, 2, 3, \dots, p-1\}$ is the
Set of all natural numbers
that are less than p and
Co-prime to p .

Q:- Which of the following is/are not true.

~~(a)~~ $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\oplus_6$
not closed $2 \oplus_6 4 = 0 \notin \text{Set}$

~~(b)~~ $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\otimes_6$
 $2 \otimes_6 3 = 0 \notin \text{Set} \therefore$ not closed

(True) (c) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$

~~(d)~~ $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\otimes_5$
 $\hookrightarrow \text{inv}(0) = \text{does not exist w.r.t. } \otimes_5$



Topic : Order of an element in a group $(G, *)$

Let $(G, *)$ be a finite group of order $= |G|$.

For an element $a \in G$,

Order of element 'a' can be denoted by $O(a)$ (or $\langle a \rangle$)

and it is defined as the least positive integer n

such that

$$(a)^n = e \text{ (identity element)}$$

→ i.e., it can not be '0'

$$(a)^n = \underbrace{a * a * a * \dots * a}_{n \text{ times 'a'}}$$

Note:

If 'e' is the identity element of the group,
then

$$(e)^1 = e \text{ (identity)}$$

∴ Order of identity element of the group
is always '1'.

eg. $\{1, -1\}$ is a finite group of order = 2 w.r.t. multiplication
find the order of every element of the group.

• $1 = \text{identity} \therefore O(1) = 1$

$$O(G) = 2$$

• $(-1)^1 = -1$

$$(-1)^2 = 1 = \text{identity} \therefore O(-1) = 2$$

$$(-1)^0 = 1$$

but 0
not allowed

eg. $\{1, \omega, \omega^2\}$ is a finite group of order=3 wrt. multiplication
find the order of every element of the group.

* $1 = \text{identity} \quad \therefore O(1) = 1$

→ $(\omega)^1 = \omega \neq 1$

$(\omega)^2 = \omega^2 \neq 1$

$(\omega)^3 = \omega^3 = 1 = \text{identity} \quad \therefore O(\omega) = 3$

* $(\omega^2)^1 = \omega^2 \neq 1$

$(\omega^2)^2 = \omega^4 = \omega \neq 1$

$(\omega^2)^3 = \omega^6 = 1 = \text{identity} \quad \therefore O(\omega^2) = 3$

$O(G) = 3$

eg. $\{1, -1, i, -i\}$ is a finite group of order = 4 wrt. multiplication
find the order of every element of the group.

• $1 = \text{identity} \therefore O(1) = 1$

• $(-1)^1 = -1 \neq 1$

$(-1)^2 = 1 = \text{identity} \therefore O(-1) = 2$

• $(i)^1 = i$

$(i)^2 = i^2 = -1$

$(i)^3 = -i$

$(i)^4 = 1 = \text{identity} \therefore O(i) = 4$

$O(G) = 4$

• $(-i)^1 = -i$

$(-i)^2 = -1$

$(-i)^3 = i$

$(-i)^4 = 1 = \text{identity} \therefore O(-i) = 4$



Topic : Properties w.r.t. Finite Group

- ① order of identity element is always '1'.
- ② For any finite group $(G, *)$.
Order of any element of group $(G, *)$ is less than or equal to the order of the group.

Lagrange's
theorem

(And)
Order of each element of group $(G, *)$ divides the order of group $(G, *)$

- ③ Order of an element $a \in$ group, will be exactly same as order of its inverse i.e., $O(a) = O(a^{-1})$



Topic : Properties w.r.t. Finite Group

- ④ Order of an element 'a' of the group can be '1' if and only if element 'a' is the identity element of the group.
- ⑤ For an element $a \in G$
 if $o(a) = 2$ { i.e. $a \times a = e(\text{identity})$ }
 then element 'a' is inverse of itself.

H.W.

Q. We know $\{0, 1, 2, 3, 4\}$ is a finite group of
Order = 5, w.r.t. binary opⁿ $\oplus_5$.

Find order of each element of the group.

Order of group is = 5, which is a prime number,

Order of each element of the group must divide '5'

∴ Order of an element of the above group can be '1' or '5'

• Order can be '1' only for identity element, $\implies 0(0) = 1$

&
for every non-identity element order must be '5' $\implies$

$$0(1) = 5$$

$$0(2) = 5$$

$$0(3) = 5$$

$$0(4) = 5$$

$\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$

* $0 = \text{identity} \Rightarrow \therefore \boxed{O(0) = 1}$

* $(1)^1 = 1$

$(1)^2 = 1 \oplus_5 1 = 2$

$(1)^3 = 1 \oplus_5 1 \oplus_5 1 = (1 \oplus_5 1) \oplus_5 1$
 $= (1)^2 \oplus_5 1$
 $= 2 \oplus_5 1 = 3$

$(1)^4 = (1)^3 \oplus_5 1 = 3 \oplus_5 1 = 4$

$(1)^5 = (1)^4 \oplus_5 1 = 4 \oplus_5 1 = 0 = \text{identity} \Rightarrow \therefore O(1) = 5$

$\hookrightarrow \text{inv}(1) = 4 \Rightarrow \therefore \boxed{O(4) = O(1) = 5}$

$(2)^1 = 2$

$(2)^2 = 2 \oplus_5 2 = 4$

$(2)^3 = (2)^2 \oplus_5 2 = 4 \oplus_5 2 = 1$

$(2)^4 = (2)^3 \oplus_5 2 = 1 \oplus_5 2 = 3$

$(2)^5 = (2)^4 \oplus_5 2 = 3 \oplus_5 2 = 0 = \text{identity}$

$\therefore \boxed{O(2) = 5}$

$\hookrightarrow \text{inv}(2) = 3$

$\therefore \boxed{O(3) = O(2) = 5}$



Topic : Subgroup

- * Let $(G, *)$ be a group, a subset H of set G is a subgroup of group $(G, *)$ only if $(H, *)$ is a group.
- * Every set is a subset of itself, so set ' G ' is a subset of set ' G ' itself and will satisfy all properties of group w.r.t. binary opⁿ $*$, therefore $(G, *)$ is always a subgroup of group $(G, *)$.
- * If ' e ' is the identity element of group $(G, *)$, then $\{e\}$ is always a subset of set G , and we know $\{e\}$ will always follow all properties of a group w.r.t. binary opⁿ ' $*$ ', therefore $(\{e\}, *)$ is always a subgroup of group $(G, *)$.

Note:

Let $(G, *)$ be a group with 'e' as its identity element. then $(G, *)$ and $(\{e\}, *)$ are always subgroups of group $(G, *)$ and they are called trivial subgroups of group $(G, *)$

Every other subgroup of group $(G, *)$ is called "proper subgroup" of group $(G, *)$

eg: We know $\{1, -1, i, -i\}$ is a group w.r.t. multiplication.

Here $\rightarrow \{1\}$ is a trivial subgroup of order = 1 w.r.t. opⁿ multiplication

$\rightarrow \{1, -1, i, -i\}$ is also a trivial subgroup of above group

And $\rightarrow \{1, -1\}$ is a proper subgroup of above group.

And order of subgroup = 2

eg. $\{1, 3, 5, 7\}$ is a group w.r.t. $\otimes_8$

• $\{1\}$ and $\{1, 3, 5, 7\}$ are trivial subgroups of above group

and $\{1, 3\}$, $\{1, 5\}$, $\{1, 7\}$ are proper sub-groups of above group.



Topic : Properties w.r.t. Subgroup

- ① Let $(G, *)$ be a group and H is any non-empty subset of set G ,
then $(H, *)$ is a subgroup of group $(G, *)$
Only if $a * b^{-1} \in H, \forall a, b \in H$



Topic : Properties w.r.t. Subgroup

② Let $(G, *)$ be a group, and $(H, *)$ be a subgroup of group $(G, *)$ such that
 $|G|$ is order of group $(G, *)$ and
 $|H|$ is order of subgroup $(H, *)$
then $|H|$ always divides $|G|$

Lagrange's
Theorem

Order of Subgroup always divides the order of original group!

Q:-

Let $(G, *)$ be a group of order $= p$.

where p is a prime number, then

How many total subgroups are possible for group $(G, *)$

Soln.

We know order of subgroup must divide the order of the group

Here order of group is ' p ' (a prime number)

there are only two divisors of p

i.e.

1

and

p

itself

it will be wrt-subgroup $(\{e\}, *)$

it will be wrt-group $(G, *)$ itself

∴ There will be only two sub-groups and both of them are 'trivial'



Topic : Properties w.r.t. Subgroup

③ Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ be two subgroups of group $(G, *)$, then $(H_1 \cap H_2, *)$ is also a subgroup of group $(G, *)$

Let $a, b \in H_1 \cap H_2$

then $a, b \in H_1$

↓
We know $(H_1, *)$
is a group (Subgroup)

$$\therefore a * b^{-1} \in H_1$$

and

$a, b \in H_2$

↓
We know $(H_2, *)$
is a group (Subgroup)

$$\therefore a * b^{-1} \in H_2$$

then

$$a * b^{-1} \in H_1 \cap H_2$$

ie. if $a, b \in H_1 \cap H_2$, then $a * b^{-1} \in H_1 \cap H_2$

Hence $H_1 \cap H_2$ will be a group



④ Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ be two subgroups of group $(G, *)$ then $(H_1 \cup H_2, *)$ is a subgroup of group $(G, *)$ if and only if $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$

$\downarrow$ $\downarrow$

i.e. $H_1 \cup H_2 = H_2$ i.e. $H_1 \cup H_2 = H_1$

Counter example

We know $\{0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \dots\}$ is a group wrt. "addition"

let $H_1 = \{0, \pm 2, \pm 4, \pm 6, \dots\}$ all multiples of 2

and $H_2 = \{0, \pm 3, \pm 6, \pm 9, \dots\}$ all multiples of 3

We can see $(H_1, +)$ and $(H_2, +)$ are sub groups of above group

$$H_1 \cup H_2 = \{0, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 9, \dots\}$$

$$\begin{array}{c} \downarrow \quad \swarrow \\ 2 + 3 = 5 \notin H_1 \cup H_2 \end{array}$$

→ is not closed wrt. binary opⁿ addition

∴ $(H_1 \cup H_2, +)$ is not a group

} in the above eg
neither $H_1 \subseteq H_2$
nor $H_2 \subseteq H_1$



Topic : Cyclic group

* Let $(G, *)$ is a group. If there exists any element $a \in G$ such that Every element of set G can be represented in the form $(a)^n$ for some positive integer 'n', then

$$(a)^n = \underbrace{a * a * a * \dots * a}_{n \text{ times } a}$$

group $(G, *)$ is called a cyclic group. and element 'a' is one of the generator of cyclic group $(G, *)$

Note: A cyclic group may have more than one generator

Note: Let $(G, *)$ be the group and 'e' is the identity element of the group.

$$(e)^1 = e$$

$$(e)^2 = e * e = e$$

$$(e)^3 = e^2 * e = e * e = e$$

$\vdots$

Identity element of the group
can not generate any other
element except itself.

∴ Identity element can never be the generator of a group containing any other element except itself.

Q: $\{1, -1\}$ is a group w.r.t. multiplication

Here $\odot$ $1 = \text{identity}$ $\therefore$ can not generate any other element except itself

$\odot \left. \begin{aligned} (-1)^1 &= -1 \\ (-1)^2 &= 1 = e \end{aligned} \right\}$ all the element of the group
can be generated by '-1'

Note: Order of group = 2
 $O(-1) = 2$

$\therefore \{1, -1\}$ w.r.t. multiplication is a
cyclic group and '-1' is the
generator of the cyclic group.

H.W.

① Check whether the group $\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group or not. If cyclic then find all the generators of the cyclic group

② Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. If cyclic then find all the generators of the cyclic group

③ Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not. If cyclic then find all the generators of the cyclic group



2 mins Summary



✓ **Topic**

Subgroup

✓ **Topic**

Cyclic group

THANK - YOU